Objective:

Test whether mannitol is effective in different enviorrments to mitigate bleaching in sea anemones

Model Organism:

Sea anemones model coral as they use the same symbiotic algae to eat

Reef Relief: The Effects of UV, Chemical, and Thermal Stressors on the Ability of the Antioxidant Mannitol to Mitigate Bleaching in the Sea Anemone *Exaiptasia pallida*

Hypothesis:

If mannitol is added to the sea anemones bleached due to elevated temperature, then it will more effectively mitigate bleaching compared to the sea anemones bleached due to UV or menthol, because mannitol targets the hydroxyl radical which may not be present in anemones bleached due to other causes.

Introduction

Coral Bleaching:
Corals contain zooxanthellae (symbiotic algae) which is their food source. **Chemical pollution**, **increased temperature, and too much sun** cause excess reactive oxygen species which cause damage to cell membranes, proteins, and DNA. To get rid of ROS, corals expel their symbionts, causing them to **lose their color and food source.**

Dasgupta, S. (2015, October 13). Massive bleaching event puts world's coral reefs at risk. Mongabay Environmental News. https://news.mongabay.com/2015/10/massive-bleaching-event-puts-worlds-coral-reefs-at-risk/

Mannitol:
A sugar alcohol used in medical research (isomer of sorbitol). It is an antioxidant shown to mitigate bleaching in sea anemones by bringing an **increase in resistance to oxidative stress.** The project "The Effect of the Antioxidant Mannitol on Bleaching in the Sea Anemone *Exaiptasia pallida*," found evidence that mannitol was effective in mitigating beaching due to elevated temperature, but not due to excess UV. This study aims to expand this idea and **test the effect of three different forms of bleaching on mannitol's effectiveness: elevated temperature, excess UV light, and the addition of menthol to model chemical pollution.**

Overview

Independent Variable: Type of Stressor (UV, Temp, Menthol)
Dependent Variable: Symbiont density
- Lower symbiont density (cell count of zooxanthellae) is an indicator of bleaching

4 Trials: Elevated temperature, excess UV light, addition of menthol (chemical stressor), control group (no stressor)

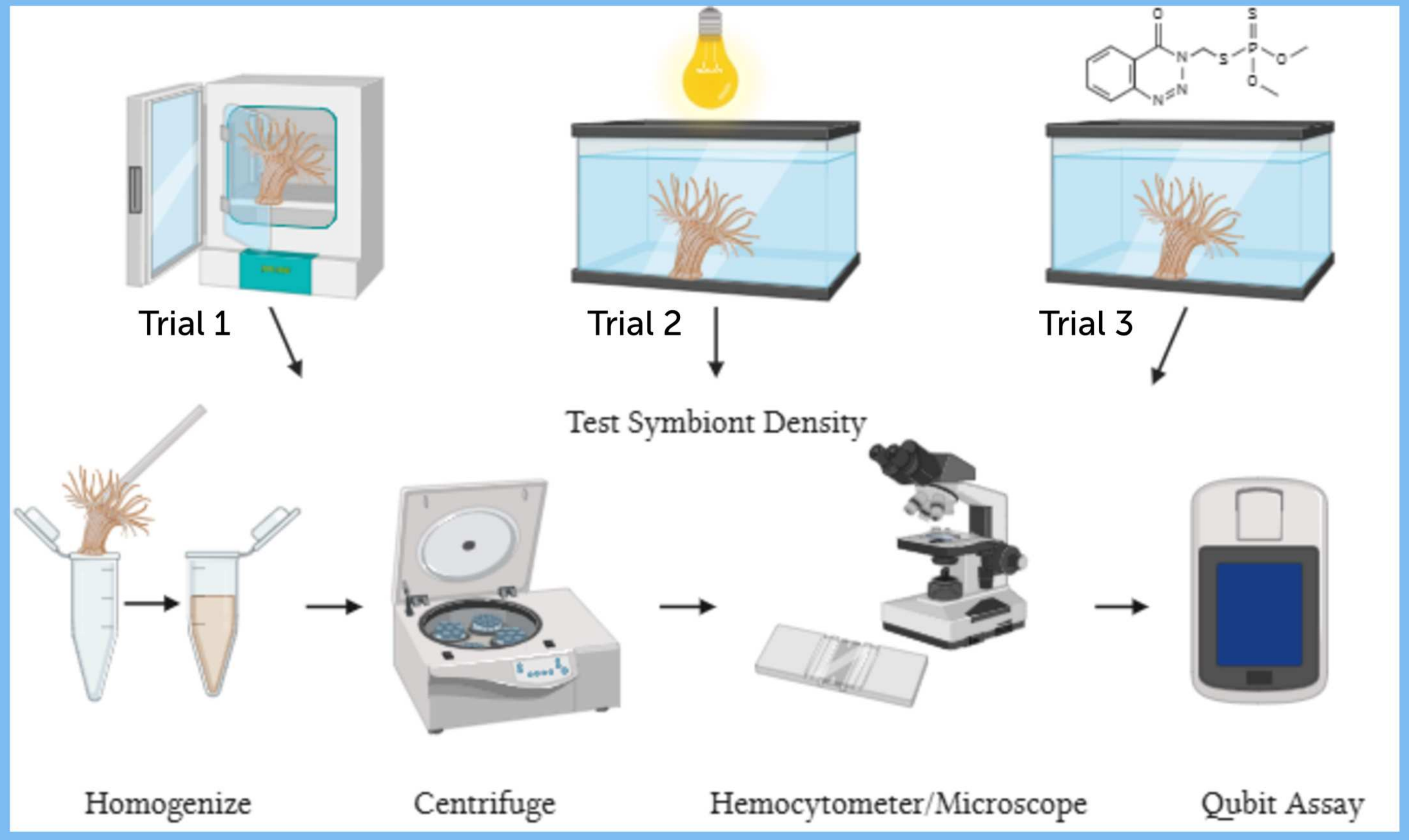
Methods

Stage 1: Acclimation
- 10 days, fed with brine shrimp

Stage 2: Mannitol and Menthol
- 1 M stock solution, add to the experimental group (10 mM)
- 10 mL 1 M stock solution of menthol, add to Trial 3 (0.15 mM)

Stage 3: Exposure
- Anemones were randomly assigned to mannitol/no mannitol group
- Trial 1 (Temp): Temp rose by 1 °C every other day (26°C-32°C)
- Trial 2 (UV): Light 16 H on, 8 H off for 2 weeks
- Trial 3 (Menthol): 0.15 mM Menthol for 7 hours, 2 days
- Trial 4 (Control): No exposure, 2 weeks

Stage 4: Symbiont Density
- Test anemones for symbiont density by counting cells with hemocytometer
- Normalize symbiont density to total protein count found with Qubit Assay



Discussion

Trial 1 (Elevated Temperature):
Although no statistical test could be run due to insufficient data, the visual trend showed that **mannitol-treated anemones retained more symbionts than the untreated group.** This pattern is consistent with the hypothesis, suggesting that mannitol may offer protection against heat-induced oxidative stress.

Trial 2 (Excess UV):
Mannitol did not mitigate bleaching caused by UV exposure (p = 0.61). This aligns with the hypothesis. UV stress often causes direct DNA damage and photochemical injury, not just oxidative stress. Because these pathways do not rely heavily on hydroxyl radicals, **mannitol's mechanism is less effective.** The data strongly supports that mannitol would not help under UV-induced bleaching.

Trial 3 (Menthol / Chemical Stress):
Mannitol significantly reduced bleaching under menthol exposure (p = 0.02). Menthol is known to disrupt cellular processes and generate reactive oxygen species, including hydroxyl radicals. Mannitol's antioxidant activity likely neutralized these ROS, helping maintain symbiont density. The menthol results support that **mannitol is also effective against chemically induced oxidative stress.**

Trial 4 (Control):
Even without any stressor, Mannitol-treated anemones had higher symbiont density (p = 0.01). This suggests that **mannitol may enhance baseline cellular stability or reduce naturally occurring oxidative stress.**

Results

Trial 1:
Not enough data points to run a statistical test due to anemone death.

Trial 2:

Degrees of Freedom (df)	U-Statistic	Z-Statistic	P-value	Interpretation of P
8	15	0.64	0.61	A P-value of 0.61 means no evidence that the groups might be different.

Trial 3:

Degrees of Freedom (df)	U-Statistic	Z-Statistic	P-value	Interpretation of P
10	33	2.4	0.02	A P-value of 0.02 means strong confidence that the groups are different.

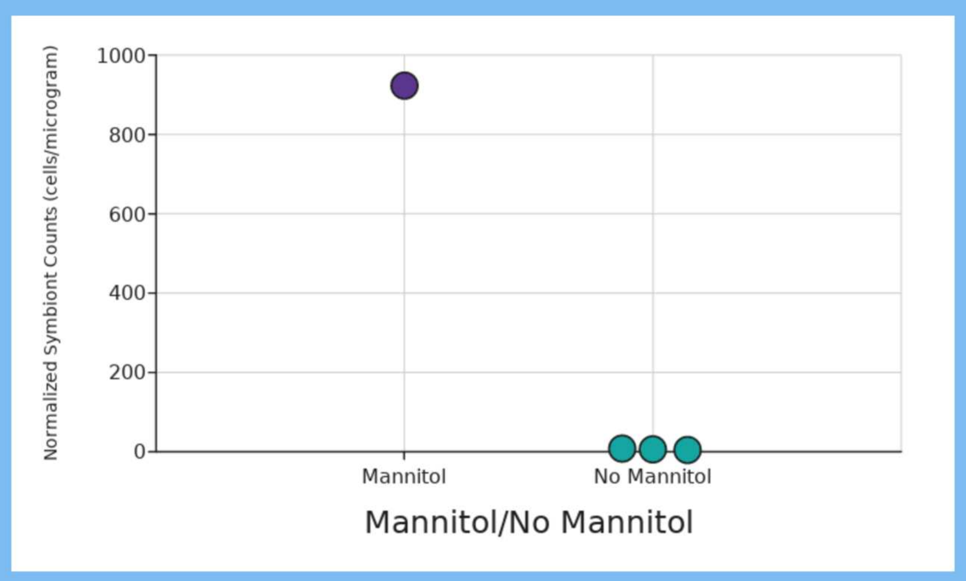
Trial 4:

Degrees of Freedom (df)	U-Statistic	Z-Statistic	P-value	Interpretation of P
8	0	-2.6	0.01	A P-value of 0.01 means very strong confidence that the groups are different.

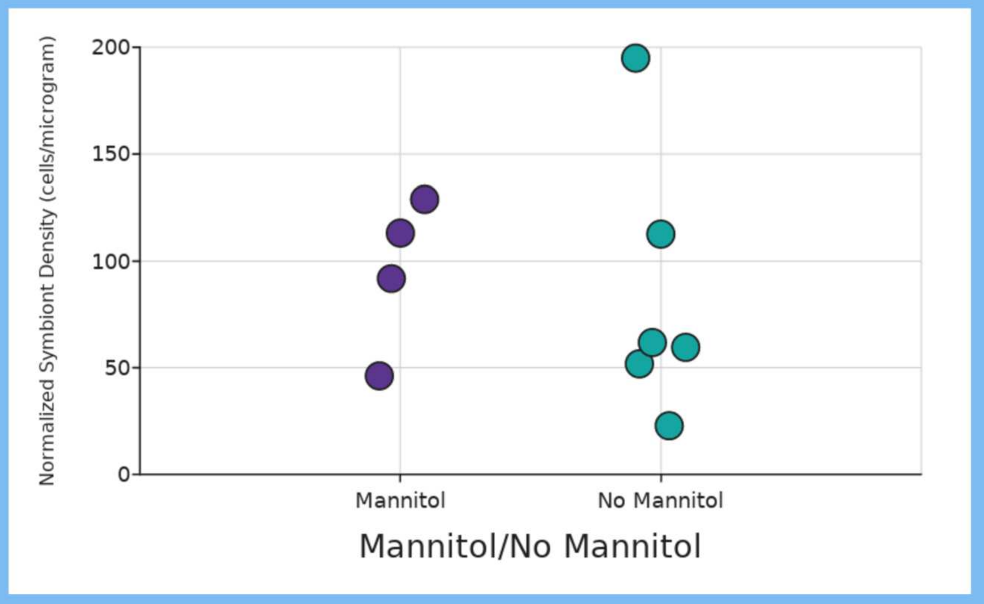
Overall:
Mannitol's effectiveness differed significantly across stressors. Statistical tests (Mann-Whitney) showed no significant protection under UV exposure (p = 0.61), while menthol stress and the control condition produced significant increases in symbiont density in mannitol-treated anemones (p = 0.02 and p = 0.01). Although the elevated-temperature trial could not be analyzed due to insufficient data, the available results indicate that mannitol is statistically ineffective against UV-induced bleaching but statistically beneficial under chemical stress and normal conditions.

Results

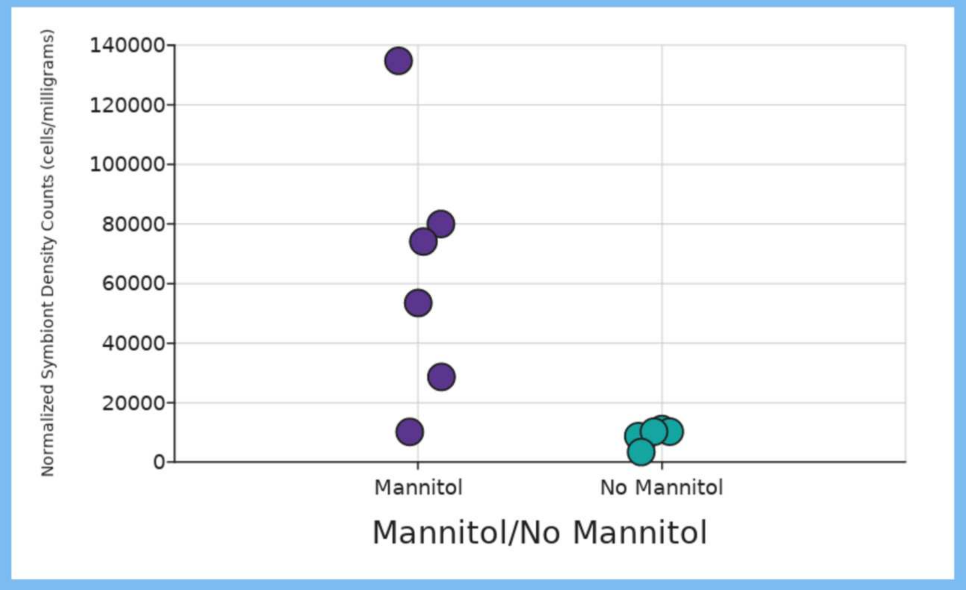
Trial 1 - Elevated Temp



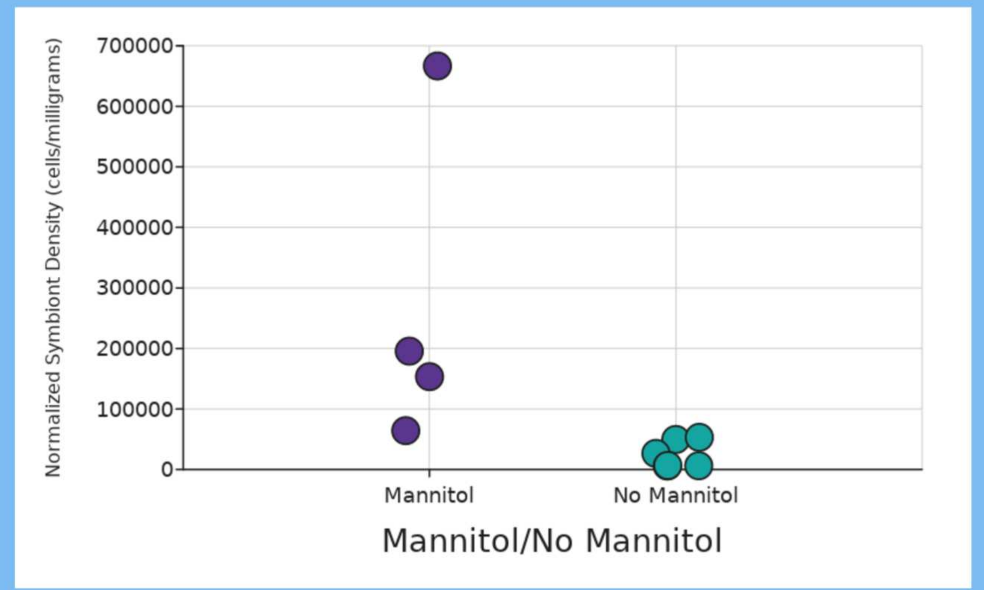
Trial 2 - Elevated UV



Trial 3 - Menthol



Trial 4 - Control



Future Work

Mannitol could help increase coral resilience, particularly in environments affected by chemical runoff or rising temperatures, where oxidative stress is a major driver of bleaching. Moreover, **mannitol's antioxidant properties may inspire the development of new treatments or supplements** designed to protect a variety of marine organisms from oxidative stress. Future studies should **investigate long-term effects of mannitol,** and how it performs under combined stress scenarios such as simultaneous UV exposure and elevated temperature.

References

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